

| Question | Answer | Mark | Mark scheme                                                                                                   | Additional guidance                                                                                                                                                                                                                    |
|----------|--------|------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18       | 30.6   | P1   | for process to find $TC$ , eg. $(TC =) 14 \times \frac{3}{3+4} (= 6)$                                         | Lengths of $TC$ , $TD$ , $SD$ may be seen on the diagram                                                                                                                                                                               |
|          |        | P1   | for process to find $TD$ , eg. $(TD =) \sqrt{14^2 + "6"^2}$ or $\sqrt{232}$ or $2\sqrt{58}$ ( $= 15.2\dots$ ) |                                                                                                                                                                                                                                        |
|          |        | P1   | for process to find $SD$ , using area of a trapezium,<br>$147 = 0.5 \times (SD + 12) \times 14$ , or $SD = 9$ |                                                                                                                                                                                                                                        |
|          |        | P1   | for $\tan^{-1}\left(\frac{"9"}{"15.2\dots"}\right)$                                                           |                                                                                                                                                                                                                                        |
|          |        | A1   | for answer in the range 30.4 to 30.7                                                                          | A complete set of processes to find the angle is needed where an alternative route is involved with more than one stage in the working<br>If an answer is given in the range in working and then rounded incorrectly award full marks. |